

# Madiou Sarr

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## Education

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### Amadou Mahtar Mbow University – Polytech Diamniadio

Diamniadio, Senegal

*B.Eng. in Computer Engineering (L2), Information Systems, Data & Security*

*Nov 2024 – Present*

- Relevant Coursework: Operating Systems Fundamentals, Database Modeling, Linux & Windows Systems Administration

### Seydou Nourou Tall Private School Group

Dakar, Sénégal

*Scientific Baccalaureate*

*2023 – 2024*

## Projects

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### SCABS SIEM | *Python, Django, Machine Learning (Scikit-learn), Chart.js*

2025

- Designed a security diagnostic platform for SMEs featuring an anomaly detection engine (Isolation Forest, Random Forest) and a SIEM correlator.
- Built an IDS rules engine (brute-force, port scan, and DoS detection) paired with an auto-defense module based on iptables/netsh.
- Implemented a SQL injection testing module, automated PDF reporting, email alerts, and an interactive Chart.js dashboard.
- Published the project on GitHub with full documentation after successful testing against an authorized target.

### FOOD DEL | *MERN Stack (MongoDB, Express.js, React, Node.js), Stripe, JWT*

2025

- Built a full-featured food delivery application with a Node.js/Express back end, a React/Vite client, and an admin panel.
- Integrated JWT authentication, MongoDB Atlas, and Stripe payment processing.
- Resolved a critical security incident (credentials exposed on GitHub) by rotating secrets and cleaning the Git history, reinforcing DevSecOps practices.

### Device Tracker | *FastAPI, SQLite, WebSocket, Progressive Web App, systemd (in progress)*

2025 – Present

- Building a full-stack geolocation tracking system as a Linux/GitHub portfolio project, deployed on Ubuntu 22.04 with systemd service management.
- Pivoted the client from React Native/Expo to a Progressive Web App after mobile build issues, restoring real-time device registration over WebSocket.

### Redis Clone in C | *C, TCP Sockets, RESP Protocol (in progress)*

2025 – Present

- Implementing a Redis-compatible server from scratch as part of the *build-your-own-x* project to deepen low-level systems and network programming skills.
- Set up a working TCP server on port 6379; currently developing a RESP protocol parser.

## Technical Skills

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**Languages:** Python, C, JavaScript, SQL, HTML/CSS

**Frameworks & Libraries:** Django, FastAPI, Node.js, Express.js, React.js, React Native

**Databases:** MongoDB, SQLite, MySQL

**Systems & Administration:** Linux (Ubuntu/Debian), Windows Server, Bash, PowerShell, systemd, VirtualBox, TCP/IP Networking

**Tools:** Git, GitHub, VS Code, Postman, GNS3, Wokwi, Tinkercad

## Leadership & Involvement

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Actively maintain a public GitHub portfolio ([github.com/Madjilouh](https://github.com/Madjilouh)) documenting complete projects from design through deployment.

Delivered a full technical report and oral presentation on Linux/Windows multi-user administration for an end-of-module systems administration case study.

Pursuing complementary certifications: CompTIA Linux+, LPIC-1, CompTIA Security+, AWS Cloud Practitioner.